

Inverse relations and functions



1 $y = x$

2

a No

b Yes

c No

d Yes

e No

f Yes

g No

h Yes

3

a Yes

b No

c Yes

d Yes

e No

f No

g Yes

h Yes

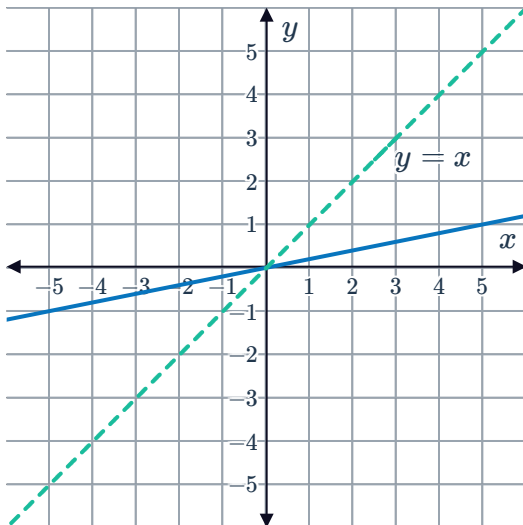
i Yes

j Yes

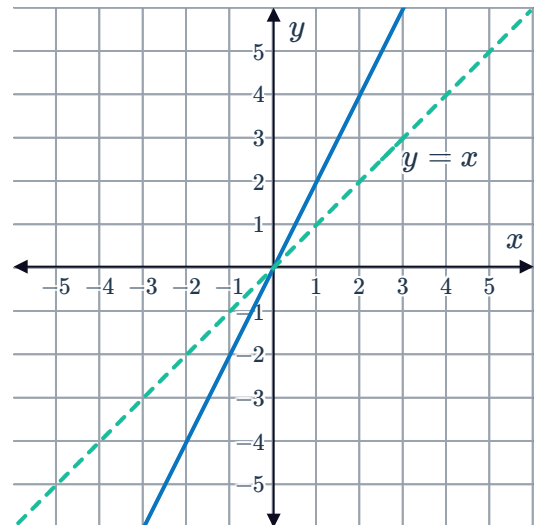
k No

4

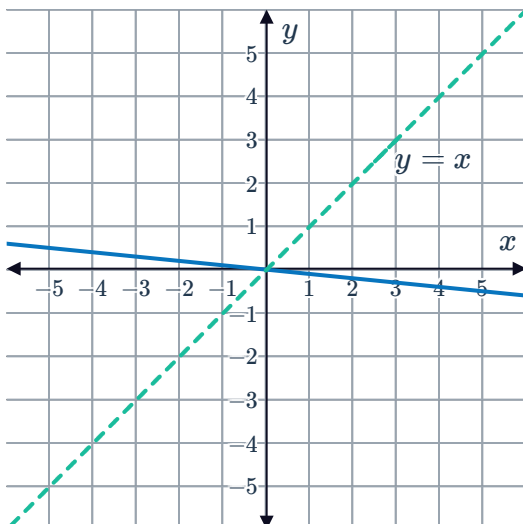
a



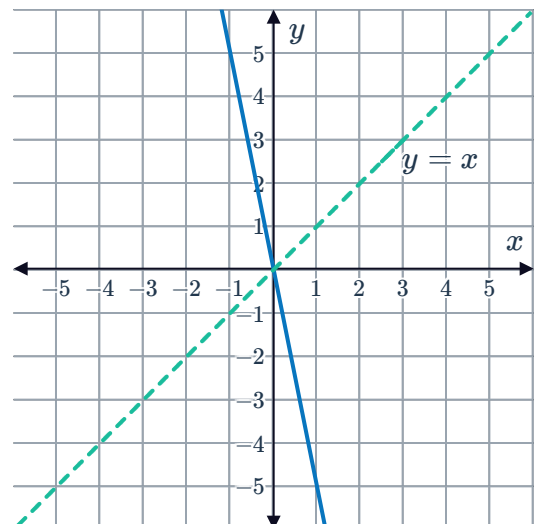
b



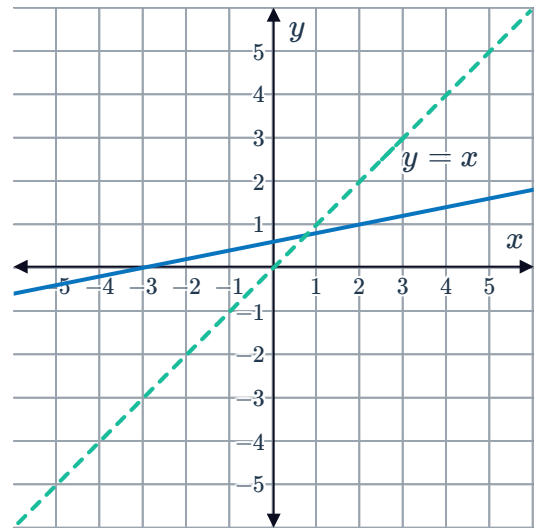
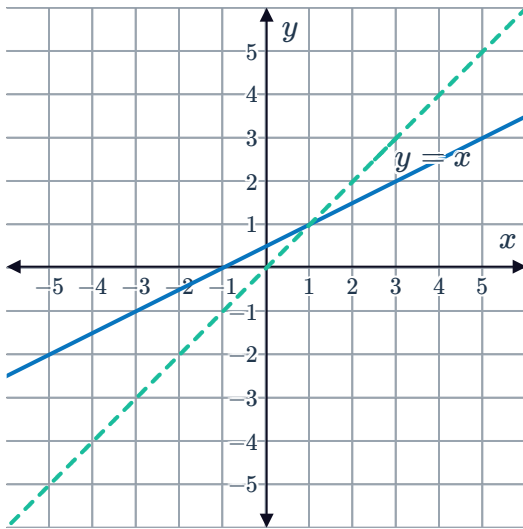
c



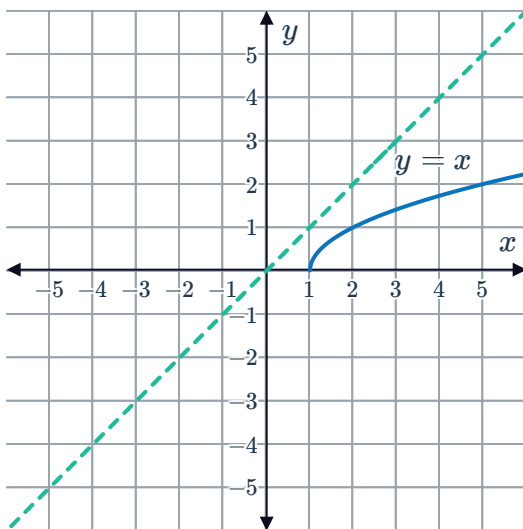
d



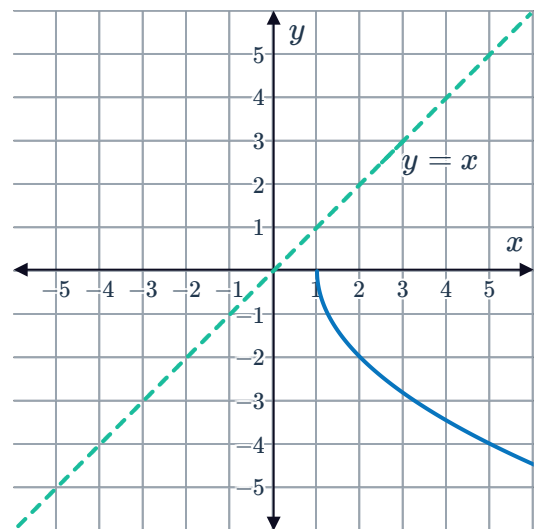
e



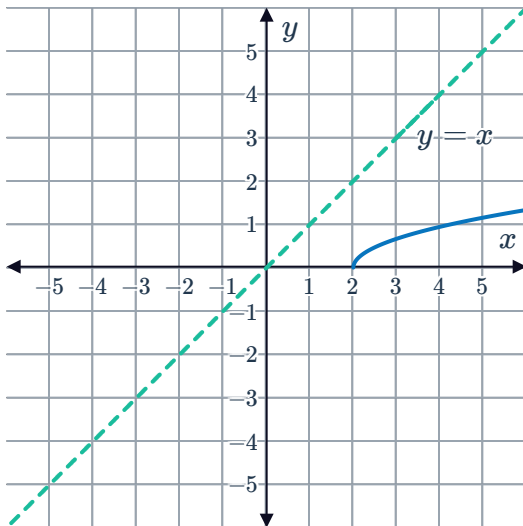
g



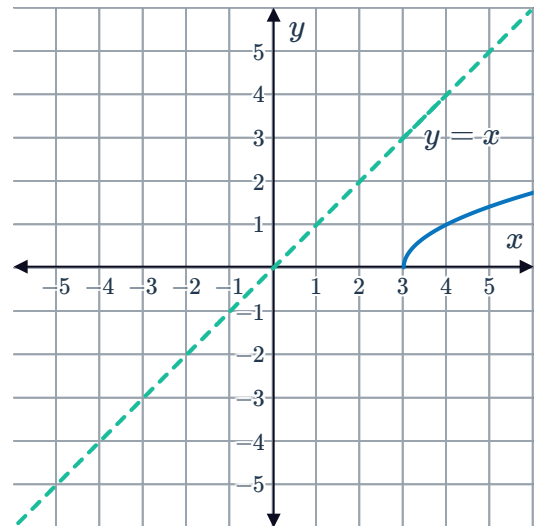
h



i



j



5

a Function



b Relation

6

a

i x

ii x

iii Yes

b

i $x - 12$

ii $x - 4$

iii No

c

i x

ii x

iii Yes

7

a

$y = \frac{x}{5}$

b

$y = \frac{x+8}{7}$

c

$y = \frac{3}{4}(x+5)$

d

$y = \sqrt[3]{\frac{x}{6}}$

e

$y = 8(1-x)$

f

$y = 6x + 7$

g

$y = x^3 + 2$

h

$y = \sqrt[3]{x-3} + 4$

i

$y = -\frac{2x}{x-5}$

j

$y = \frac{8}{x-5} + 9$

k

$y = \frac{8}{5x} - \frac{3}{5}$

l

$y = \sqrt[3]{\frac{7}{x+3}} + 8$

m

$y = \frac{\sqrt[3]{\frac{x+7}{4}} + 2}{9}$

Domain and range

8

a $\{-3\}$

b $\{6\}$

c $(6, -3)$

d $\{6\}$

e $\{-3\}$

f The domain of f is equal to the range of f^{-1} , and the range of f is equal to the domain of f^{-1} .

9

a $\{6, -5\}$



b $\{6, -5\}$

10

a $\{5, 7, 8\}$

b $\{9, -4, 2\}$

c $\{(9, 5), (-4, 7), (2, 8)\}$

d $\{9, -4, 2\}$

e $\{5, 7, 8\}$

11

a $a = -\frac{1}{2}, b = \frac{5}{2}$

b $[5, 0]$

c $[-5, 5]$

12

a

i $f^{-1}(x) = \frac{1}{5}(x - 15)$

ii $(-\infty, \infty)$

iii $(-\infty, \infty)$

b

i $f^{-1}(x) = \frac{1}{3}(4 - x)$

ii $(-\infty, \infty)$

iii $(-\infty, \infty)$

c

i $f^{-1}(x) = \frac{1}{8}(x + 9)$

ii $[-41, 7]$

iii $[-4, 2]$

d

i $f^{-1}(x) = \sqrt{x}$

ii $[0, \infty)$

iii $[0, \infty)$

e

i $f^{-1}(x) = \sqrt{16 - x^2}$

ii $[0, 4]$

iii $[0, 4]$

13

a

i $f(x) = \frac{2x}{3} - 2$

ii $g(x) = \frac{3x}{2} + 3$

b

i x

ii x

c

i True

ii True

iii False

iv True



14

a $[-4, 9]$

b $[-4, 5]$

c $[-4, 5]$

d $[-4, 9]$

e False

15

a

i $(0, 1)$

ii $(1, 2)$

iii $(2, 4)$

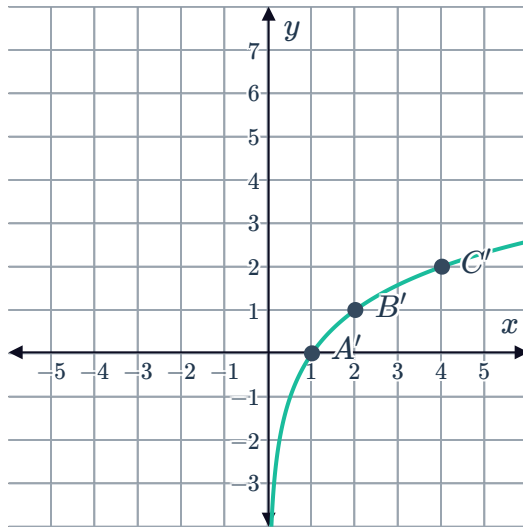
b

i $(1, 0)$

ii $(2, 1)$

iii $(4, 2)$

c



16

a $(-\infty, \infty)$

b $(0, \infty)$

c $(0, \infty)$

d $(-\infty, \infty)$

e The range of f^{-1}

f The domain of f^{-1}

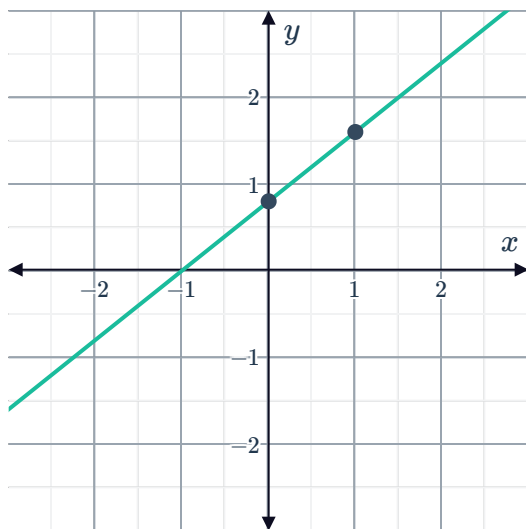
17

a $x \geq 2$

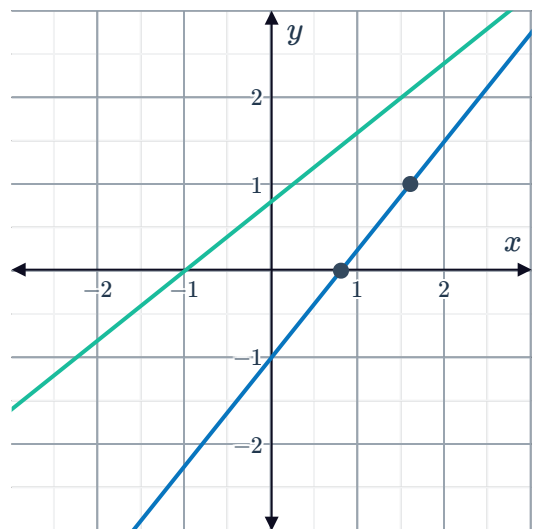
b $y \geq -5$

c $y = x^2 - 4x + 9$

a

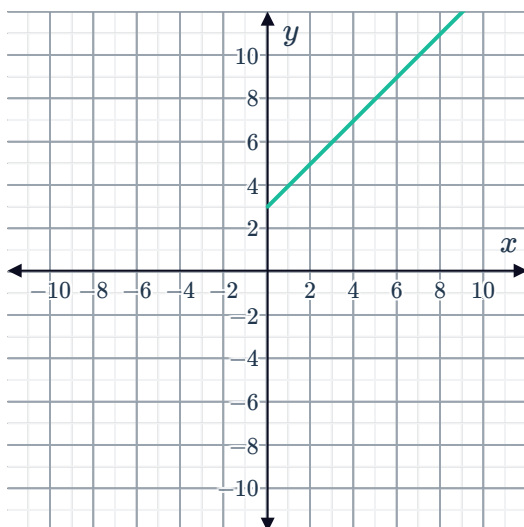


b

c $(-\infty, \infty)$ e $(-\infty, \infty)$ d $(-\infty, \infty)$ f $(-\infty, \infty)$

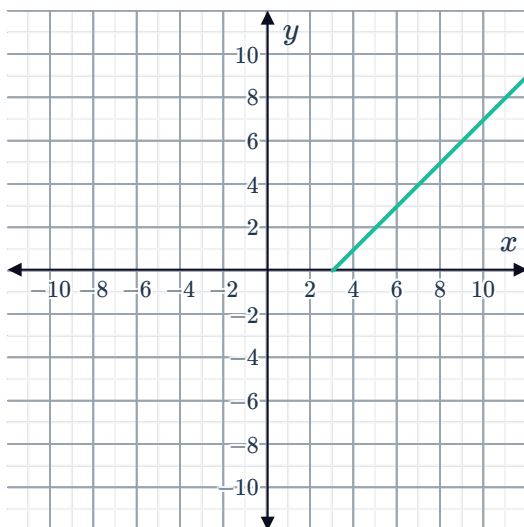


ii $f^{-1}(x) = x - 3$



iii $[3, \infty)$

iv $[0, \infty)$

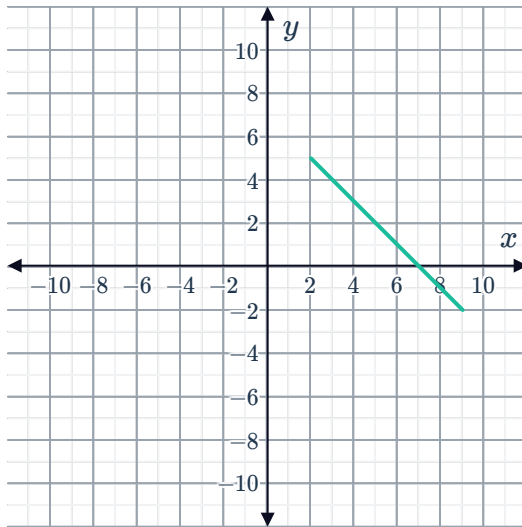


b
i



ii

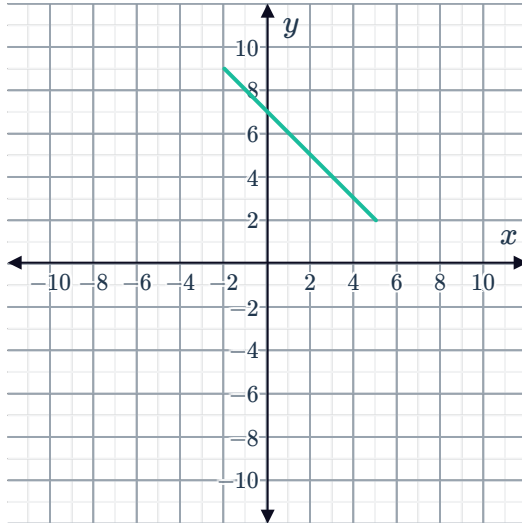
$$f^{-1}(x) = 7 - x$$



iii $[-2, 5]$

iv $[2, 9]$

v

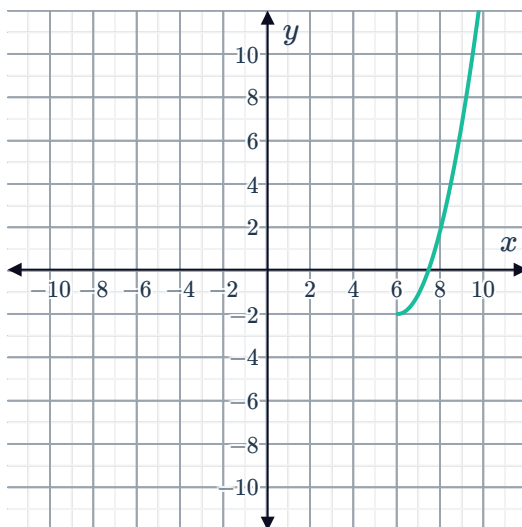


c
i



ii

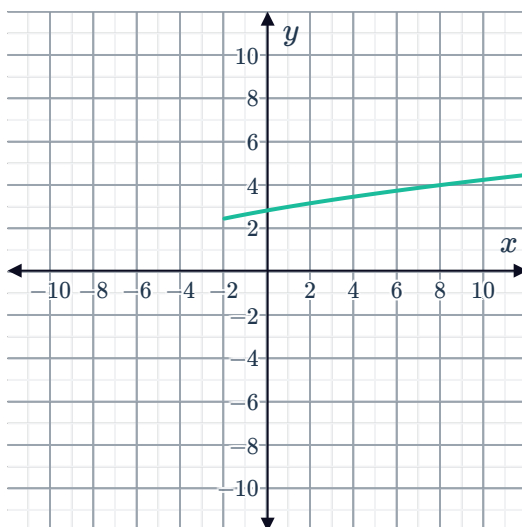
$$f^{-1}(x) = \sqrt{x+2} + 6$$



iii $[-2, \infty)$

iv $[6, \infty)$

v

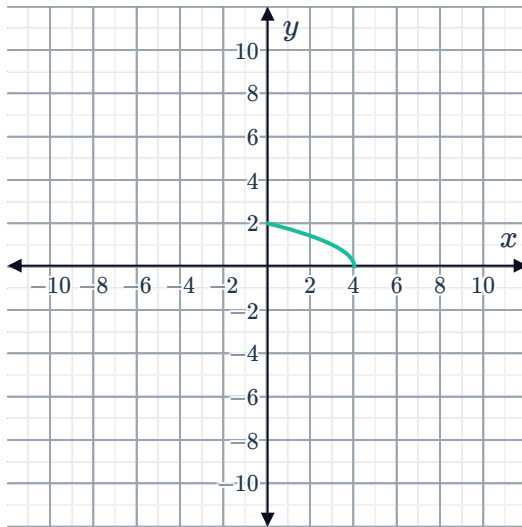


d
i



ii

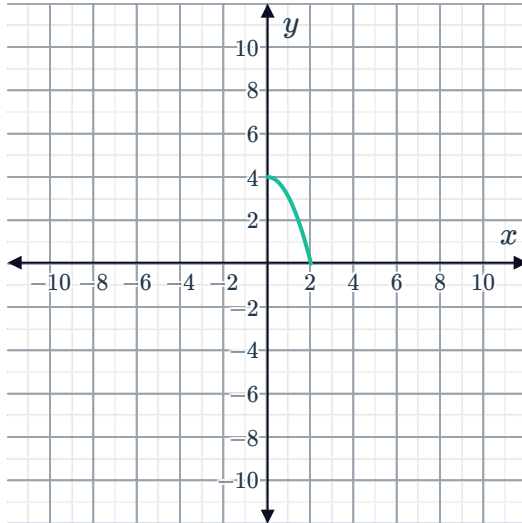
$$f^{-1}(x) = 4 - x^2$$



iii $(0, 2]$

iv $[0, 4)$

v



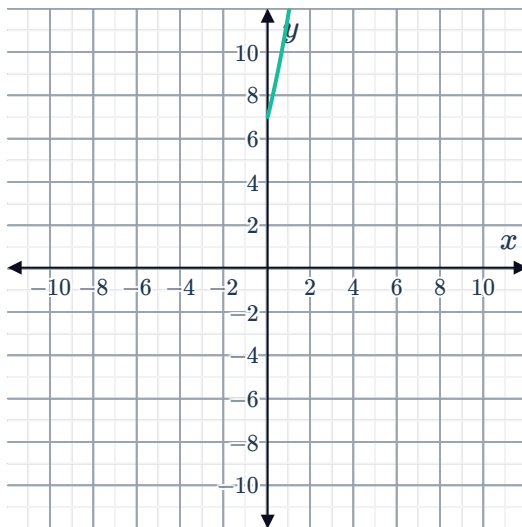
e

i



ii

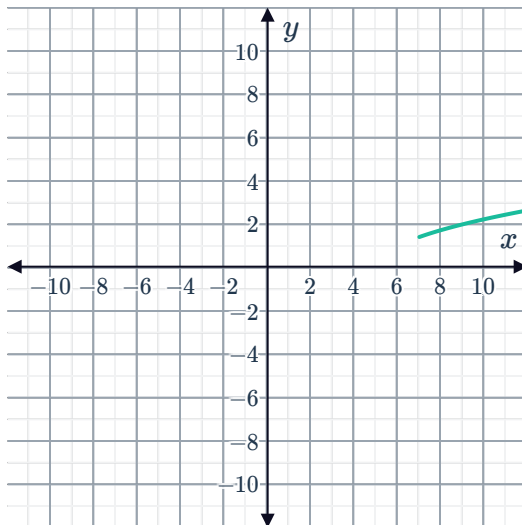
$$f^{-1}(x) = \sqrt{x-3} - 2$$



iii $[7, \infty)$

iv $[0, \infty)$

v



20

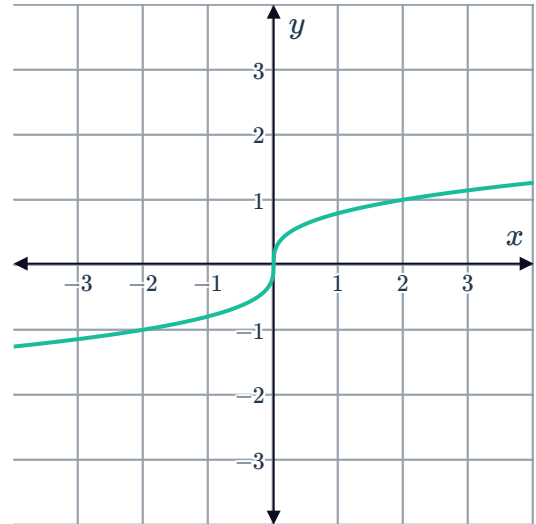
a Yes

c All real y



b All real x

d



21

a No

c $x > 2$ and $x < -2$

b All real x

d $y > 2$ or $y < -2$

22

a $T = 1.07A$

b $A = \frac{T}{1.07}$